

$^{35}\text{Cl}(\text{n}, \text{n}'\gamma)$ 1989Ge09, 1984El12, 1966Ni03

1984El12, 1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two ^{35}Cl targets were placed at 45° and 135° with a distance of 70 cm apart. γ rays were detected using a Ge(Li) detector. Measured E_γ . Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of CCl_4 and a 5.6-cm-diam sphere of C_2Cl_6 . γ rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets. γ rays were detected using a Ge(Li) detector. Measured 2 γ rays and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>J^π#</u>	<u>T_{1/2}[‡]</u>
0	3/2 ⁺	
1219	1/2	139 fs 56
1762	5/2	
2646	7/2	187 fs 62
2694		43 fs 6
3002		50 fs 9
3163		

[†] From a least-squares fit to γ -ray energies.

[‡] From 1984El12 and 1989Ge09 using DSAM.

From $\gamma(\theta)$ in 1966Ni03, except for the ground state from the Adopted Levels.

 $\gamma(^{35}\text{Cl})$

<u>E_γ</u>	<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
1219	1219	1/2	0	3/2 ⁺	
1762	1762	5/2	0	3/2 ⁺	E _γ : from 1966Ni03. A ₀ =+0.942 8, A ₂ =-0.148 23 (1966Ni03).
2646	2646	7/2	0	3/2 ⁺	A ₀ =+1.078 5, A ₂ =+0.165 12 (1966Ni03).
2694	2694		0	3/2 ⁺	
3002	3002		0	3/2 ⁺	
3163	3163		0	3/2 ⁺	E _γ : from 1966Ni03.

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Level Scheme

